Comparative Evaluation of Deep Learning Models for Binary Sentiment Classification on the IMDB Movie Review Dataset

Synopsis/Interim/Final report on
Deep Learning Project
[ICT-4442]



Submitted By
ARNAV KUMAR | 230911308

SCHOOL OF COMPUTER ENGINEERING
MANIPAL INSTITUTE OF TECHNOLOGY,
MANIPAL ACADEMY OF HIGHER EDUCATION

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PART A : SYNOPSIS TEMPLATE (Phase 1, due Tue, 25 Aug 2026)

Fill in every field below. Keep the synopsis to 1–2 pages.

Team No. and Names (with Registration Numbers):

[YOUR TEAM NO.] | ARNAV KUMAR | 230911308

Title of the Project:

Comparative Evaluation of Deep Learning Models for Binary Sentiment Classification on the IMDB Movie Review Dataset

Brief Description of the Project (not more than 500 words):

The automated classification of text into either positive or negative sentiment is one of the fundamental problems in NLP that serves as the foundation for many downstream applications such as mining customer product reviews and social media sentiment. In this work, I will examine the effect of increasing complexity in terms of inductive biases on the performance of the four deep learning architectures on the binary classification benchmark IMDB Large Movie Review Dataset. Starting from the classical MLP Bag-of-Words (BoW) model, where reviews are represented by either fixed-length TF-IDF or multi-hot vectors and sent to the dense layers, thus disregarding the word order to create a strong non-sequential baseline. Next, I employ 1D Convolutional Neural Network (TextCNN) over the pretrained GloVe embeddings with several different kernel sizes (2, 3, 4 tokens). This architecture is able to capture the local patterns of the sentiment-bearing phrases in the form of n-grams. Bi-directional LSTM (BiLSTM) with attention layer is used to model the relations between words in terms of their order, which is vital for detecting the cases when sentiment in the review is negated. Lastly, fine-tuned DistilBERT Transformer encoder pre-trained on large text corpus is a state-of-the-practice attention-based architecture for text classification. The four models have been evaluated in the same manner: the same preprocessing (lowercase, punctuation removal, tokenisation, truncation of the vocabulary to 10,000 tokens, padding and truncating reviews to 512 tokens), 70/15/15 train/validation/test split, and evaluation metrics (accuracy, precision, recall, F1 score).

Dataset(s) Identified (with details of the dataset):

Name: IMDB Large Movie Review Dataset Source: Stanford AI Lab (available via tensorflow.keras.datasets.imdb and HuggingFace datasets library) Details: 50,000 English movie reviews labelled as positive or negative sentiment. Balanced split: 25,000 training and 25,000 test samples (12,500 positive and 12,500 negative each). Reviews vary in length from a few sentences to several paragraphs (average ~231 words). No missing labels. Already tokenised version available via Keras for rapid prototyping; raw text version available via HuggingFace for custom preprocessing pipelines.

Proposed Models for Comparison (3–4, spanning different architecture families):

Classical Baseline - MLP w/ TF-IDF BoW: Dense net over fixed-size TF-IDF feature vector representations. No information about the ordering of words; sets the performance floor with no consideration for sequentiality. CNN Variant - TextCNN: 1D convolutions over GloVe 300 word embeddings with kernel sizes 2, 3, 4; max over time pooling, and a sigmoid classifier. RNN/LSTM Variant - BiLSTM w/ Attention: Stacked layers of bidirectional LSTMs over GloVe word embeddings with an attention mechanism to find out words that determine the sentiment. Attention-based - DistilBERT (fine-tuned): Fine-tuned HuggingFace DistilBERT base-uncased for binary classification task.

Model-to-Member Assignment (one model/task per member):

ARNAV KUMAR (230911308) — All four models: MLP BoW Baseline, TextCNN, BiLSTM with Attention, and DistilBERT fine-tuning, shared preprocessing pipeline, evaluation harness, and comparative analysis.

Existing State-of-the-Art Method(s) Related to the Proposed Project:

BERT (Devlin et al., 2019): Bidirectional Encoder Representations from Transformers achieved near-human performance on IMDB (>95% accuracy) via masked language model pre-training and downstream fine-tuning. RoBERTa: Robustly optimised BERT with dynamic masking and larger batch sizes; further improved BERT-level IMDB accuracy. ULMFiT (Howard & Ruder, 2018): Universal Language Model Fine-Tuning

demonstrated that language model pre-training with discriminative learning rates substantially outperforms training from scratch on sentiment tasks. TextCNN (Kim, 2014): Showed that simple convolutional architectures with pre-trained word vectors are competitive with recurrent models on sentence-level classification tasks, including sentiment.

Objectives (crisp and clear, not more than 2–3; must not describe activities/steps):

1. Evaluate the performance of MLP BoW, TextCNN, BiLSTM-Attention, and DistilBERT architectures in terms of binary sentiment classification accuracy, F1-score, and inference latency on the IMDB dataset using a common evaluation protocol. 2. To measure the performance gap between training-from-scratch models (MLP, CNN, BiLSTM) and fine-tuned pre-trained transformers (DistilBERT) on a large balanced binary text classification benchmark and to find the cost-benefit trade-off in terms of training time and parameter count.

Importance / Motivation of the Proposed Project:

Sentiment analysis on products and entertainment reviews underpins multi-billion dollar consumer decision infrastructure. IMDB is the gold standard for binary sentiment analysis, meaning comparisons against this data set will have direct comparisons to results from decades of literature. Four distinct architectures range in performance from the least compute-intensive (MLP Bag-of-Words — CPU, seconds per epoch) to the most powerful (DistilBERT — GPU recommended, minutes per epoch). This is an example of how the Pareto front between accuracy and compute power works in practice. In the case of applications like embedded review analysers or on-device natural language processing, knowing when TextCNN will achieve 90% of the accuracy of DistilBERT with 1% of the parameter count is extremely relevant. This assignment touches upon all major NLP architecture families covered in ICT-4442.

Preliminary Reference List (minimum 5 papers):

[1] A. L. Maas et al., “Learning Word Vectors for Sentiment Analysis,” in Proc. ACL, pp. 142–150, 2011. [2] Y. Kim, “Convolutional Neural Networks for Sentence Classification,” in Proc. EMNLP, pp. 1746–1751, 2014. [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in Proc. NAACL, pp. 4171–4186, 2019. [4] J. Howard and S. Ruder, “Universal Language Model Fine-Tuning for Text Classification,” in Proc. ACL, pp. 328–339, 2018. [5] V. Sanh et al., “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” arXiv:1910.01108, 2019. [6] Z. Yang et al., “Hierarchical Attention Networks for Document Classification,” in Proc. NAACL, pp. 1480–1489, 2016.

Work Plan:

Sl. No.	Component / Work Element / Milestone	Expected Completion (Day/Month)
1	IMDB dataset loading (Keras and HuggingFace), EDA (review length distribution, vocabulary size), shared preprocessing pipeline (tokenisation, padding, TF-IDF vectors)	28/08
2	MLP BoW baseline and TextCNN implementation with GloVe-300 embeddings; training and preliminary accuracy/F1	06/09
3	BiLSTM with attention implementation and training; DistilBERT fine-tuning setup (HuggingFace Trainer)	18/09
4	Hyperparameter tuning, per-model error analysis (misclassified review examples), training time benchmarking, comparative results table, and interim report	25/09
5	Final report write-up, code cleanup and documentation, GitHub repository with commit history, presentation preparation	25/10